



OBJECT ORIENTED SOFTWARE ENGINEERING

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Software Engineering

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The IEEE definition:

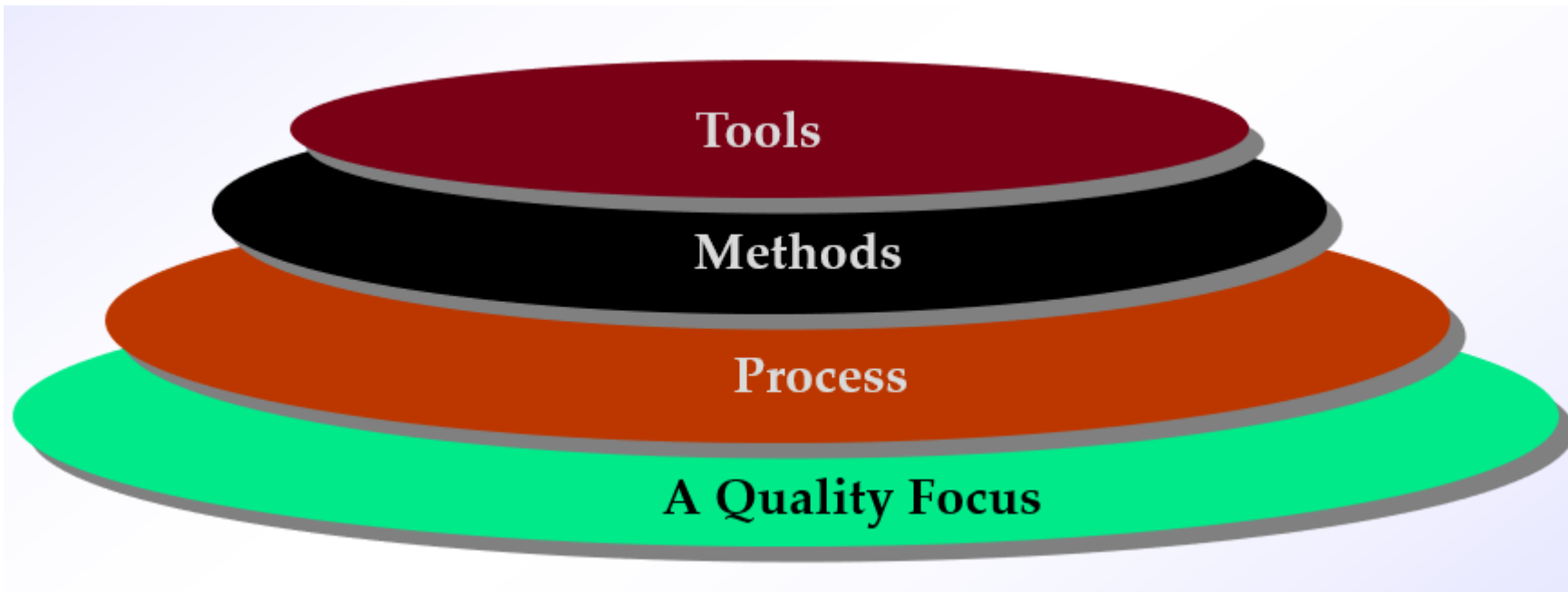
Software Engineering: (1) The application of *a systematic, disciplined, quantifiable approach* to the *development, operation, and maintenance* of software; that is, the application of engineering to software.

(2) The study of approaches as in (1).

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Software Engineering Layers

- Software Engineering is a **layered Technology**



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The Software Process

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- A process is a collection of **activities, actions and tasks** that are performed when some work product is to be created.
- **An activity** try towards achieving the broad objective regardless of specific domain.
- **Ex: communication with stakeholders – gathering requirements from different categories of users** (need not be all technically knowledgeable)

An *action* *encompasses* a **set of tasks** that produce a major work product.

- **Ex—Building an architectural design model – which includes various set of tasks in it.**

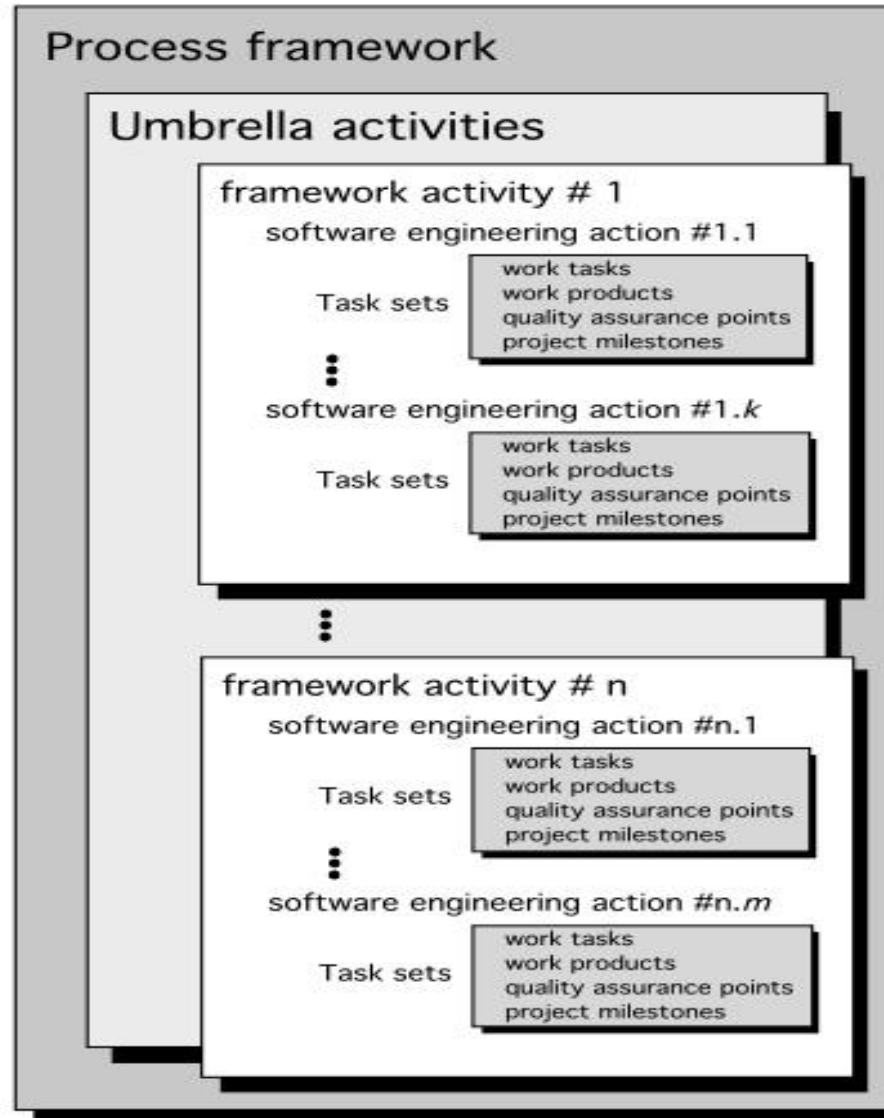
A *task* *focuses on a small, but well-defined objective* that produces a tangible outcome

- **Ex--*conducting a unit test* — which includes micro level testing**

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The Generic Process Model

Software process



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Process Framework

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- A *process framework* establishes the foundation for a complete software engineering process by :
 - identifying a small number of *framework activities* that are applicable to all software projects and
 - a set of *umbrella activities* that are applicable across the entire software process.

A **generic process framework** for software engineering encompasses **five** activities:

1. **Communication**

2. **Planning**

3. **Modeling**

Analysis of requirements,
Design

4. **Construction**

Code generation
Testing

5. **Deployment**

1. Communication

- Before any technical work can commence, it is important to **communicate and collaborate with the customer.**
- The intent is **to understand** stakeholders' objectives about the project .
- To **gather requirements** that helps to define software features and functions.

2. Planning

- **Any complicated journey can be simplified if a **map exists**.**
In software project the planning activity creates a “map” that helps to **guide the team**, makes the journey.
- The map—called a ***software project plan***—***defines the software engineering work by describing***
the **technical tasks** to be conducted,
the **risks** that are likely,
the **resources** that will be required,
the **work products** to be produced, and
a **work schedule**.

3. Modeling

- Create a “sketch” of the thing so that one can understand the big picture.
 - In other sectors, a landscaper, a bridge builder, an aeronautical engineer, a carpenter, or an architect, is the one who perform this (modeling) activity.
- Here, a **software engineer** does the same thing by **creating models** to **better understand** software requirements, and the design that will achieve those requirements.

4. Construction

- This activity **combines**
 - **code generation** and
 - **testing** - that is required to **uncover errors** in the code

5. Deployment

- The software is **delivered** to the **customer** who **evaluates** the delivered product and provides **feedback** based on the evaluation.



THANK YOU

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